NOTE 10 - INBOUND VS OUTBOUND TRAFFIC

Network traffic is described as inbound or outbound relative to a boundary - normally the organisation's perimeter or the host being examined. The direction of a connection is one of the first things an analyst checks, because it says a great deal about who initiated the conversation and what stage the attack has reached.

Definitions

	Inbound traffic	Outbound traffic
Direction	From outside the protected network inwards	From inside the protected network outwards
Who initiates	An external host or an attacker	An internal host, user or process
Typical examples	Someone browsing your public web server; a VPN client connecting; a scan against your public IPs; an attacker's exploit hitting a service	A workstation browsing the web; an update check; email being sent; a server uploading a backup; a C2 beacon
Normal control on the firewall	Blocked unless a service is deliberately published	Allowed but usually filtered (proxy, allow-list, category blocking)
Main security risk	Exposed services, exploits, brute force, denial of service	Data exfiltration, command and control, downloads of malicious payloads, beaconing
Typical visibility	Firewall / web-server / IDS logs on the DMZ edge	Proxy, DNS, firewall and EDR network logs

The same connection seen from two sides

A browser on `10.0.5.24` opening `https://shop.example.com` (`203.0.113.45`) is:

- Outbound from the internal workstation: `10.0.5.24:51844 -> 203.0.113.45:443`.
- Inbound at the web server: `203.0.113.45:443 <- 10.0.5.24:51844`.

So "inbound" and "outbound" are relative to a boundary. In ATT&CK terms, identifying the initiating side and the direction of the flow is what separates Initial Access (inbound to the victim) from Command and Control or Exfiltration (outbound from the victim).

A simple decision table used during triage

Observation	Likely meaning	Typical ATT&CK tactic
Inbound to a published service, normal volume, known geographies	Expected business traffic	-
Inbound to a published service from thousands of IPs, repeated failures	Scanning or brute force	Reconnaissance / Credential Access
Inbound to a port that should not be open (RDP 3389, SMB 445, SSH 22)	Misconfiguration or attempted intrusion	Initial Access
Inbound to an internet-facing web app with encoded traversal strings	Web exploitation attempt	Initial Access
Outbound to a reputable service on 443, normal volume	Normal user activity	-
Outbound to an unknown IP/host, small and regular, on 443/80/53	Possible C2 beaconing	Command and Control

Observation	Likely meaning	Typical ATT&CK tactic
Outbound large uploads to an unusual destination, off-hours	Possible data exfiltration	Exfiltration
Outbound connection immediately after a suspicious process start	Download of a payload or second stage	Execution / C2
Outbound DNS with long random subdomains at high volume	DNS tunnelling	C2 / Exfiltration
Inbound and outbound traffic in an internal-only path (host to host)	Lateral movement inside the network, i.e. east-west	Lateral Movement

How a SOC analyst uses the distinction

1. Determine who started it. Check the firewall/proxy log for the SYN direction and the first packet. A "server" that initiates a connection to an internet host is far more suspicious than a server answering one.
2. Match the destination to expectations. Outbound traffic is only meaningful in context: is the destination an approved service, a known CDN, a threat-intel match, or a brand-new domain registered last week?
3. Look at volume and timing. Human browsing is bursty; a beacon is small, regular and machine-like (for example every 60 seconds). Large outbound transfers at 03:00 are a different problem from the same size at 11:00.
4. Correlate direction with the endpoint story. Combine the direction with process creation, parent-child process, and user context from EDR: `winword.exe -> powershell.exe -> outbound 443` is a complete narrative; the outbound connection alone is not.
5. Decide the response. Inbound attack attempts on a decoy or blocked port are usually logged and closed. A confirmed outbound C2 channel means contain the host - block the destination and isolate the endpoint, because the attacker already has a foothold.
6. Feed the firewall team the facts. Effective blocking rules need destination IP/domain, port, protocol and direction - which is exactly the information this distinction produces.

Inbound and outbound in the log sources a SOC reads

Source	What it shows best
Perimeter firewall (ingress)	Inbound scans, exploit attempts, denied connections, allowed inbound to published services
Perimeter firewall (egress)	Outbound connections to the internet, policy violations, blocked C2
Proxy / secure web gateway	Full URLs and hosts for outbound HTTP/HTTPS, user attribution, data volume uploads/downloads
DNS logs	Nearly all outbound activity starts here - the name queried before the connection
EDR network events	Which process on the endpoint opened the inbound or outbound connection
NetFlow / IPFIX	Direction, volume and frequency of flows across the network - the backbone of beacon detection
IDS/IPS (e.g. Snort)	Signature and behaviour hits on inbound attack traffic and outbound C2 patterns

Key points to remember

- Inbound = into the network, usually attacker-initiated; outbound = out of the network, usually internal-host-initiated. Both are relative to a boundary.
- Inbound mostly means attempted compromise; outbound after a compromise mostly means C2 or exfiltration - so outbound anomalies are often the more urgent.
- Always combine direction with volume, timing, destination reputation and endpoint context.
- North-south traffic crosses the perimeter; east-west traffic moves between internal hosts and is where lateral movement is found.
- A block is only useful when it includes direction, IP/domain, port and protocol.